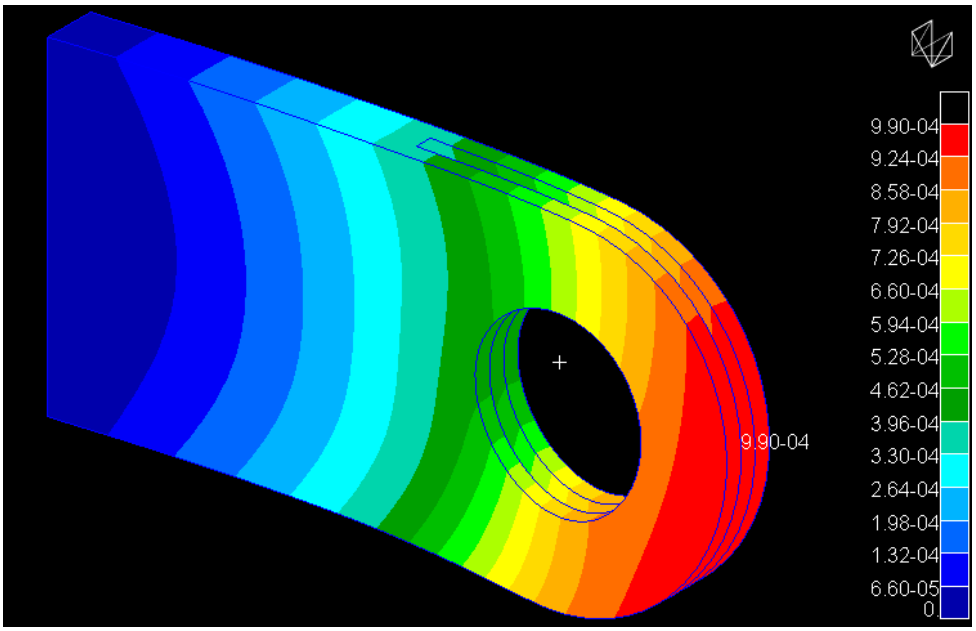
ELS2 - Mé422

NUMERICAL ANALYSIS IN STRUCTURAL ENGINEERING

TP 3 : ANALYSIS OF A SOLID LUG UNDER BENDING



Corentin PLANCHON
Teacher : LARBI Walid
Date : March 18, 2026

Contents

1 Introduction **2**
1.1 Introduction . . . 2
1.2 Geometry . . . 2
1.3 Mesh . . . 2
1.4 Material . . . 3
1.5 Boundary Conditions and Loading . . . 3

2 Results **4**
2.1 HEX8 Mesh . . . 4
2.2 HEX20 Mesh (Reference) . . . 5
2.3 TET4 Mesh . . . 6
2.4 TET10 Mesh . . . 7

3 Mesh Comparison **8**
3.1 Comparison between TET4 and TET10 . . . 8
3.2 The Influence of Mesh Density . . . 8
3.3 Error Analysis . . . 8

4 Axial Pressure Loading **9**
4.1 HEX20 Mesh . . . 9

5 Verification of the Structural Design **10**
5.1 Bending Case . . . 10
5.2 Axial Pressure Case . . . 10

6 Mesh Convergence Study **11**

7 Conclusion **12**

1 Introduction

1.1 Introduction

The objective of this work is to analyse the mechanical behaviour of a clevis subjected to a static load using the finite element method.

Because of the complex geometry of the component and the presence of a hole where the load is applied, the stress distribution cannot be obtained easily using analytical methods. A three–dimensional finite element model is therefore required.

The goal of this study is to investigate the influence of the element formulation and the mesh type on the numerical solution.

Four meshes are tested:

- HEX8 : first order hexahedral elements
- HEX20 : second order hexahedral elements
- TET4 : first order tetrahedral elements
- TET10 : second order tetrahedral elements

The HEX20 mesh is considered as the reference solution because quadratic hexahedral elements generally provide the best accuracy.

1.2 Geometry

The studied component is a mechanical clevis used in structural assemblies.

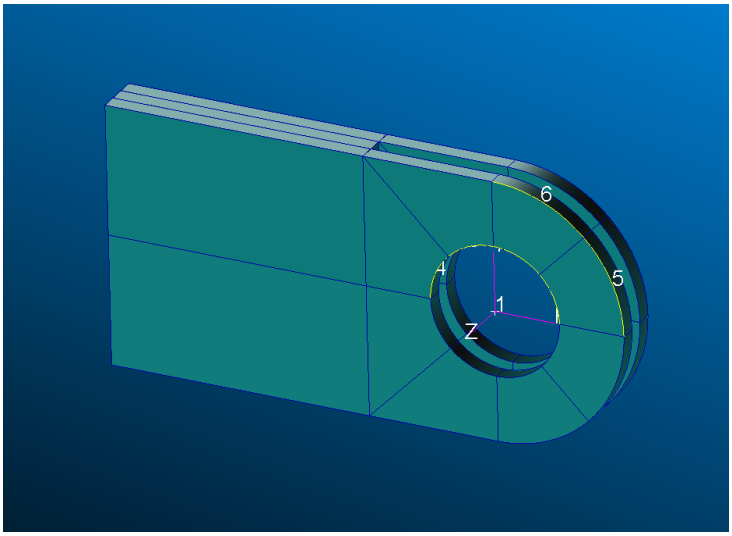


Figure 1: Geometry of the clevis

The hole introduces a geometric discontinuity that generates a stress concentration when the structure is loaded.

1.3 Mesh

The finite element mesh was generated in Patran.

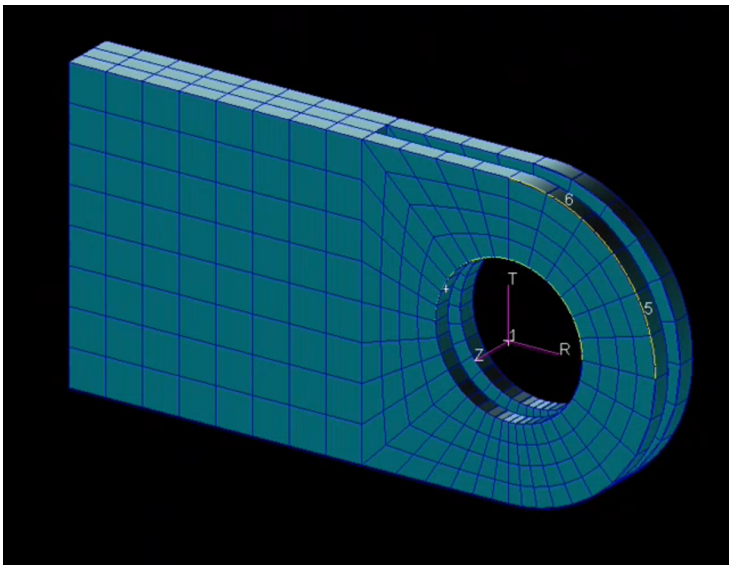


Figure 2: Finite element mesh

The mesh is refined around the hole because this region experiences the highest stress gradients.

1.4 Material

The structure is made of steel with the following properties:

$$E = 210\,GPa$$

$$\nu = 0.3$$

$$\sigma_{yield} = 235\,MPa$$

1.5 Boundary Conditions and Loading

The left face of the clevis is fully constrained:

$$T_x = T_y = T_z = 0$$

$$R_x = R_y = R_z = 0$$

The load is applied on the lower half of the hole according to the function:

$$F(x) = 100(1 - x^2)$$

This loading generates bending in the structure.

2 Results

2.1 HEX8 Mesh

Based on the generated model, it can be seen that the HEX8 mesh is a structured mesh of hexahedral elements, which have a fairly regular distribution of elements throughout the entire geometry. This area is carefully meshed to account for the expected stress concentrations in this particular area of interest. the mesh has approximately:

- 1360 nodes
- 456 elements

This mesh can be considered appropriate for a first analysis, as it remains fairly affordable in terms of computational cost. However, as HEX8 elements are linear, it is expected that the solution might remain slightly too stiff.

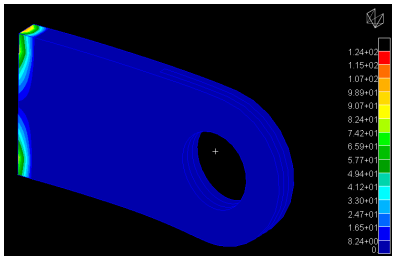


Figure 3: Reaction forces

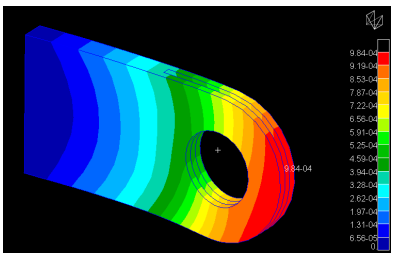


Figure 4: Displacement

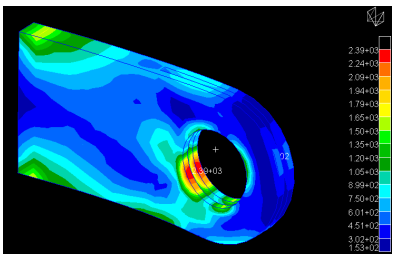


Figure 5: Von Mises stress

Figure 6: Results obtained with HEX8 elements

The maximum displacement obtained is:

$$U_{max} = 9.84 \times 10^{-4}\, m$$

The maximum Von Mises stress is:

$$\sigma_{max} = 23.9\, MPa$$

The results obtained for the given problem using the HEX8 mesh show a displacement field that is consistent with the bending behavior of the structure. The deformation increases gradually along the structure toward the hole where the loading is applied.

The stress distribution also shows the concentration of stress around the hole and the fixed support. These are the critical points of the structure. This is also expected because of the discontinuity and the boundary conditions. However, the use of the HEX8 elements for the solution of the problem also shows the stiffening behavior of the structure because of the linear interpolation used. Therefore, the maximum stress obtained is slightly less than the actual.

2.2 HEX20 Mesh (Reference)

The HEX20 mesh is also structured and regular but is based on quadratic hexahedral elements. This type of mesh is better for interpolating the displacement field and is more appropriate for calculating the stresses in the vicinity of the hole. the mesh contains approximately:

- 4267 nodes
- 456 elements

Although the number of elements is smaller compared to the HEX8 mesh, the formulation is more accurate due to the higher-order elements. This is the reason why the HEX20 mesh is generally considered to be one of the best options in cases where a structured hexahedral mesh can be generated.

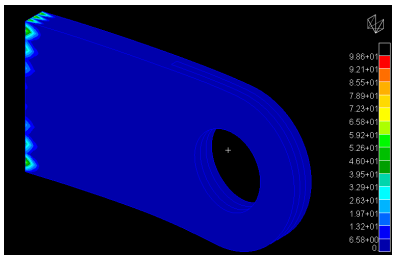


Figure 7: Reaction forces

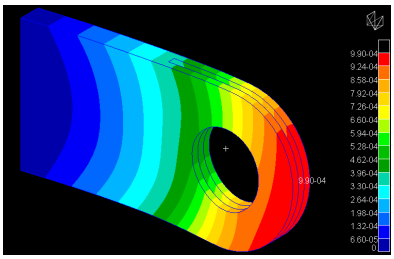


Figure 8: Displacement

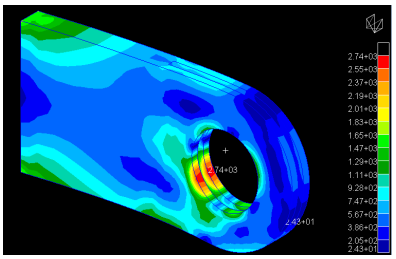


Figure 9: Von Mises stress

Figure 10: Results obtained with HEX20 elements

The maximum displacement obtained is:

$$U_{max} = 9.90 \times 10^{-4}\, m$$

The maximum stress obtained is:

$$\sigma_{max} = 27.4\, MPa$$

The results obtained by HEX20 have a smoother and more realistic distribution of both fields. The pattern of deformation is still similar to that obtained in the HEX8 case. The results obtained for the stress concentration are more accurate, where gradients are high. This is due to the quadratic interpolation used by the HEX20 elements.

Therefore, this mesh will be used as a reference solution in this study.

2.3 TET4 Mesh

The TET4 mesh is based on a significant number of small elements with a tetrahedral structure throughout the entire domain. The TET4 mesh is much less structured than the hexahedral mesh but is easier to generate automatically for complex domains. the mesh contains approximately:

- 847 nodes
- 2701 elements

The large number of elements indicates that a significant refinement is required when using first-order tetrahedral elements. Indeed, the TET4 elements are known for being stiff. In other words, a coarse mesh for the tetrahedral elements would not be enough to represent the stress field.

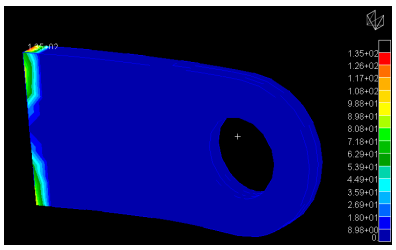


Figure 11: Reaction forces

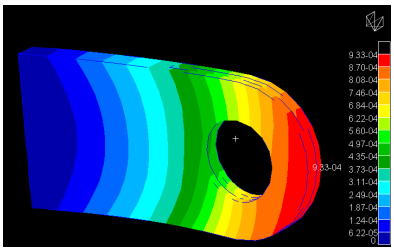


Figure 12: Displacement

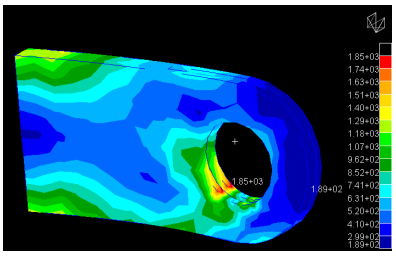


Figure 13: Stress

Figure 14: Results obtained with TET4 elements

The maximum displacement obtained is:

$$U_{max} = 9.33 \times 10^{-4}\, m$$

The maximum stress obtained is:

$$\sigma_{max} = 18.5\, MPa$$

The results obtained in the TET4 mesh show a displacement field which behaves as expected in terms of bending. However, it can be seen that the results obtained in terms of stress are not as accurate as those obtained in hexahedral meshes.

The point of maximum stress is again at the point where there is a hole, but it can be seen that this maximum stress is much lower than that obtained in the reference solution. This is due to the fact that TET4 elements tend to underestimate stress.

TET4 elements are first-order elements which are known to be numerically stiff, requiring a very fine mesh to achieve accurate results.

2.4 TET10 Mesh

The TET10 mesh is also based on tetrahedral elements, but with quadratic interpolation. It remains more flexible than a structured hexahedral mesh while improving the quality of the approximation. the mesh contains approximately:

- 5092 nodes
- 2710 elements

Compared to TET4, the number of elements is much smaller, but the solution is generally more accurate because each element is richer in interpolation. This makes TET10 a good compromise between meshing flexibility and numerical precision.

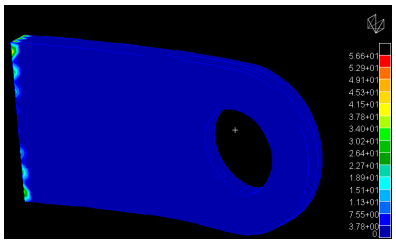


Figure 15: Reaction forces

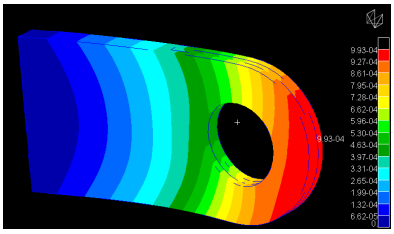


Figure 16: Displacement

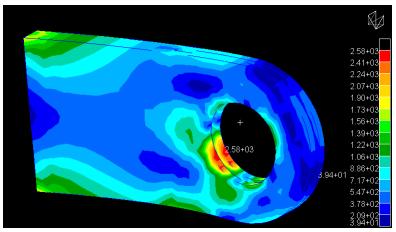


Figure 17: Stress

Figure 18: Results obtained with TET10 elements

The maximum displacement obtained is:

$$U_{max} = 9.93 \times 10^{-4}\, m$$

The maximum stress obtained is:

$$\sigma_{max} = 25.8\, MPa$$

The TET10 mesh is a much better approximation of the displacement and stress fields than the TET4 elements.

The stress field is smoother and closer to the reference solution provided by the HEX20 elements. The location of the maximum stress is again close to the hole, and this is a consistent result.

The quadratic interpolation of the TET10 elements is a good compromise between the flexibility of the meshing and the results.

3 Mesh Comparison

Mesh	Nodes	Elements	U_{max} (m)	U_{max} (mm)	σ_{max} (MPa)
HEX8	1360	456	9.84×10^{-4}	0.984	23.9
HEX20	4267	456	9.90×10^{-4}	0.990	27.4
TET4	847	2701	9.33×10^{-4}	0.933	18.5
TET10	5092	2710	9.93×10^{-4}	0.993	25.8

3.1 Comparison between TET4 and TET10

The difference between TET4 and TET10 is mainly related to the interpolation order. TET4 is a first-order element. The variation of the displacement field is linear inside an element. This is a limitation, especially for curved geometries. TET10 is a second-order element. Thanks to the mid-side nodes, there is a quadratic variation of the displacement field, which is better suited to describe stress, especially in highly loaded zones. SO TET10 is more realistic compared to TET4.

3.2 The Influence of Mesh Density

Mesh density also has a direct influence on the quality of the results from the finite element analysis. A coarse mesh may even smooth the stress field and lead to an underestimate of the peak stresses. On the contrary, a refined mesh will lead to better modeling of local stress gradients and thus yield better results. In this region, high stress concentrations are expected.

3.3 Error Analysis

The HEX20 mesh is used as the reference solution.

$$Error = \frac{|Value - Value_{ref}|}{Value_{ref}} \times 100$$

Mesh	Displacement Error (%)	Stress Error (%)
HEX8	0.61	12.77
TET4	5.76	32.48
TET10	0.30	5.84

The results clearly show the influence of the element formulation on the numerical solution.

Linear elements such as HEX8 and TET4 tend to artificially stiffen the structure, which leads to an underestimation of the stress values.

Quadratic elements such as HEX20 and TET10 provide a much better representation of the deformation field.

Among all meshes, the TET10 mesh gives results very close to the reference HEX20 solution.

4 Axial Pressure Loading

In this last part of the study, the structure is subjected to an axial pressure applied on the inner surface of the hole and is defined by the following expression:

$$p(\theta) = 700\cos(\theta)$$

This loading represents a distributed pressure acting along the circumference of the hole.

The objective of this simulation is to evaluate the stress distribution generated by this pressure and to analyse the influence of the element formulation on the numerical results.

4.1 HEX20 Mesh

A simulation was performed using quadratic hexahedra elements (HEX20).

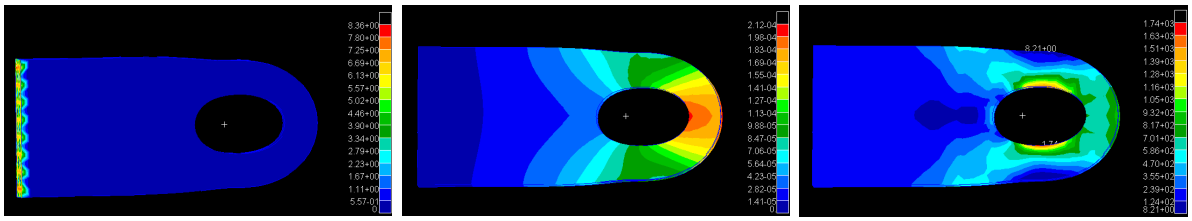


Figure 19: Reaction forces Figure 20: Displacement Figure 21: Stress

Figure 22: Results obtained with TET10 elements under axial pressure

The stress distribution is smoother and more realistic than the one obtained with TET4 elements.

The maximum stress obtained is approximately:

$$\sigma_{max} \approx 17.4 \times 10^3\, N/cm^2$$

which corresponds to:

$$\sigma_{max} \approx 17.4\, MPa$$

It can be observed that the stress field is relatively smooth, indicating that the HEX20 elements are correctly capturing the variation of the pressure along the inner surface of the hole. The maximum stress is occurring in the region where the pressure is maximum, i.e., in the direction of the cosine distribution.

It can be observed that the stresses are significantly lower compared to the case of bending. This is expected, as the stresses are occurring due to the surface loading, which is distributed, as opposed to point loading, causing a concentration of stresses.

The displacement field remains relatively small and follows the direction of the applied pressure, with maximum values located near the loaded region.

5 Verification of the Structural Design

In this section, the maximum Von Mises stress obtained for each loading case is compared to the yield strength of the material in order to evaluate whether the structure is correctly dimensioned.

5.1 Bending Case

For the bending loading, the maximum Von Mises stress obtained with the reference mesh (HEX20) is:

$$\sigma_{max}^{bending} = 27.4\,MPa$$

The yield strength of the material is:

$$\sigma_{yield} = 235\,MPa$$

Since:

$$\sigma_{max}^{bending} \ll \sigma_{yield}$$

the structure remains well within the elastic domain.

The safety factor can be estimated as:

$$SF = \frac{235}{27.4} \approx 8.6$$

This high safety factor indicates that the structure is largely over-dimensioned for the bending load.

Therefore, the part can be considered correctly dimensioned for the bending case.

5.2 Axial Pressure Case

For the axial pressure loading, the maximum Von Mises stress obtained is:

$$\sigma_{max}^{pressure} \approx 17.4\,MPa$$

Again, this value is much lower than the yield strength:

$$\sigma_{max}^{pressure} \ll \sigma_{yield}$$

The corresponding safety factor is:

$$SF \approx \frac{235}{17.4} \approx 13.5$$

This result shows that the structure is even less stressed under axial pressure than under bending.

Therefore, the part is also correctly dimensioned for the axial pressure loading.

6 Mesh Convergence Study

Mesh	Nodes	Elements	U_{max} (m)	U_{max} (mm)	σ_{max} (MPa)
HEX8	1360	456	9.84×10^{-4}	0.984	23.9
HEX20	4267	456	9.90×10^{-4}	0.990	27.4
TET4	847	2701	9.33×10^{-4}	0.933	18.5
TET10	5092	2710	9.93×10^{-4}	0.993	25.8

The influence of mesh density and element formulation is an important part of finite element analysis.

For this study, several meshes with varying element types were compared. For this purpose, HEX20 mesh has been considered as a reference solution due to its high accuracy, as it is a structured mesh with quadratic interpolation.

From this study, it has been observed that coarse meshes such as HEX8 and TET4 tend to produce a lower value of maximum stress. This is due to their low ability to approximate stress gradients, especially around the hole where a high amount of stress is concentrated.

On the other hand, high-order elements such as HEX20 and TET10 can more accurately predict the stress field. It has also been observed that results obtained using TET10 mesh are very close to reference solution results, which indicates a high convergence of results.

From this study, it can be seen that a high-order mesh is required to obtain accurate results, especially quadratic elements to predict a high amount of stress concentrations in a complex geometry.

7 Conclusion

The goal of this study was to investigate the mechanical response of a clevis piece subjected to bending using the finite element method, and to determine the impact of the mesh type and formulation of finite elements on the results.

As shown by comparing all the results, linear finite elements such as HEX8 and TET4 have a tendency to stifle the structure, which leads to a lower displacement and stress, especially in zones where there is a high stress gradient.

Conversely, quadratic finite elements such as HEX20 and TET10 provide a more realistic description of the mechanical response of the clevis piece. Among these elements, HEX20 is considered to be the best option, while TET10 is a good compromise between precision and meshing flexibility.

As far as the mesh convergence study is concerned, it is worth noticing that increasing both the mesh density and order of finite element interpolation greatly contributes to a better quality of results, especially concerning the stress concentration around the hole.

With regard to the structural response, it is worth noticing that the maximum Von Mises stress computed for both loading scenarios is much lower than the yield stress of the material, which demonstrates that the clevis piece is highly over-dimensioned.